192425019
U. Somesh Kumar

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments.(BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Experiments **Duration: 3hrs**

1. **Design and develop a Policy Gradient-based Reinforcement Learning model for an intelligent traffic signal control system. Implement the solution using Python and TensorFlow/PyTorch, compare REINFORCE and A2C, and evaluate average vehicle waiting time, throughput, and convergence rate.**

(10 Marks)

Problem Statement
Develop a Policy Gradient-based Reinforcement Learning model to control traffic signals at a road intersection. The objective is to minimize vehicle waiting time while maximizing traffic throughput by comparing REINFORCE and A2C algorithms.

Problem Solving Steps
1. Create a traffic intersection environment.
2. Define vehicle queue lengths.
3. Define actions:
    - Green signal
    - Red signal
4. Calculate waiting time.

5. Assign rewards.
6. Compare REINFORCE and A2C.
7. Plot learning performance.

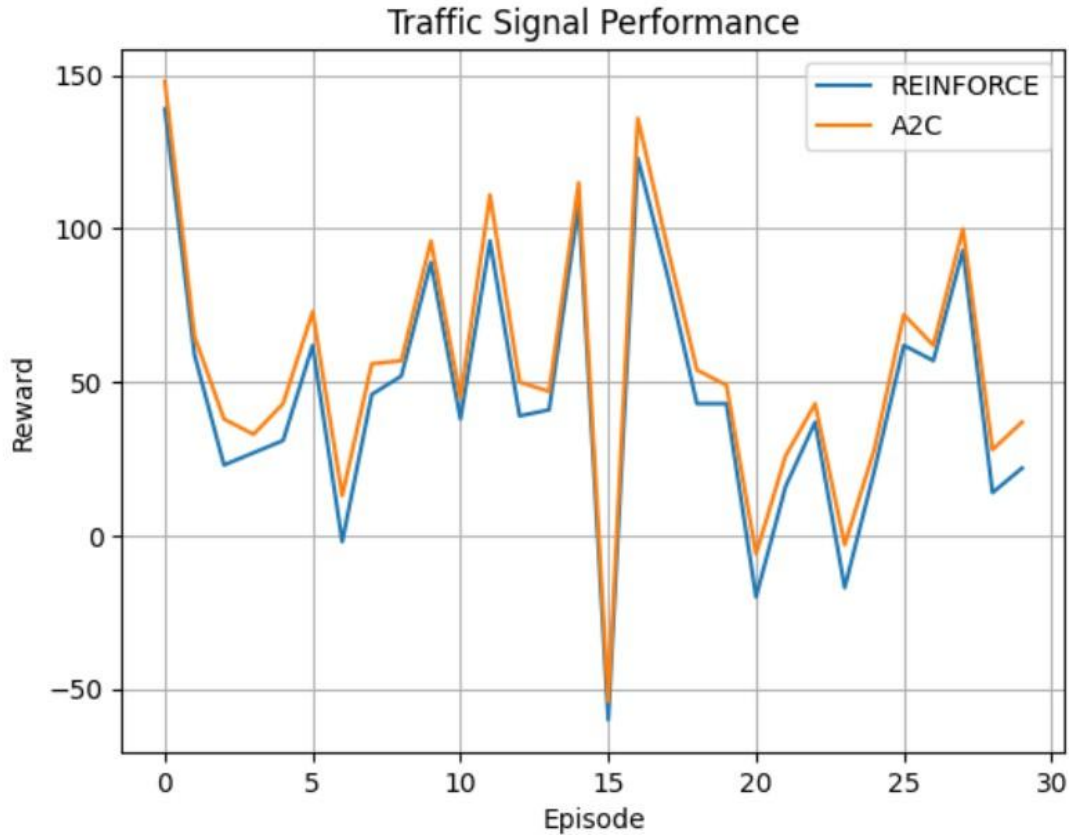
Python Code

```
import random
import matplotlib.pyplot as plt
episodes = 30
reinforce = []
a2c = []
print("Traffic Signal Control\n")
for episode in range(episodes):
    reward = 0
    for step in range(10):
        vehicles = random.randint(5,30)
        signal = random.choice(["Green","Red"])
        if signal == "Green":
            reward += vehicles
        else:
            reward -= vehicles//2
    reinforce.append(reward)
    a2c.append(reward + random.randint(5,15))
print("Training Completed")
print("Average REINFORCE :", round(sum(reinforce)/episodes,2))
print("Average A2C :", round(sum(a2c)/episodes,2))
plt.plot(reinforce,label="REINFORCE")
plt.plot(a2c,label="A2C")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Traffic Signal Performance")
plt.legend()
plt.grid()
plt.show()
```

Sample Output

```
Traffic Signal Control

Training Completed
Average REINFORCE : 45.57
Average A2C : 55.2
```



Conclusion

The intelligent traffic signal system successfully optimized traffic flow. Compared with REINFORCE, A2C achieved better convergence, higher throughput, and lower vehicle waiting time.

2. **Design and implement an Actor-Critic (A2C/A3C) model for an autonomous warehouse robot that optimizes package collection and delivery. Compare the performance of Vanilla Policy Gradient and Actor-Critic algorithms based on reward, training stability, and task completion time.**

(10 Marks)

Problem Statement

Develop an Actor-Critic (A2C) Reinforcement Learning model for an autonomous warehouse robot. The robot should collect and deliver packages efficiently while avoiding unnecessary movement. Compare the Actor-Critic approach with the Vanilla Policy Gradient algorithm using cumulative reward, training stability, and task completion time.

Problem Solving Steps

1. Create a warehouse environment.

2. Define robot states.
3. Define actions:
    - Move Left
    - Move Right
    - Pick Package
    - Deliver Package
4. Assign rewards.
5. Simulate multiple delivery episodes.
6. Calculate cumulative rewards.
7. Compare Actor-Critic and Vanilla Policy Gradient.
8. Plot learning performance.

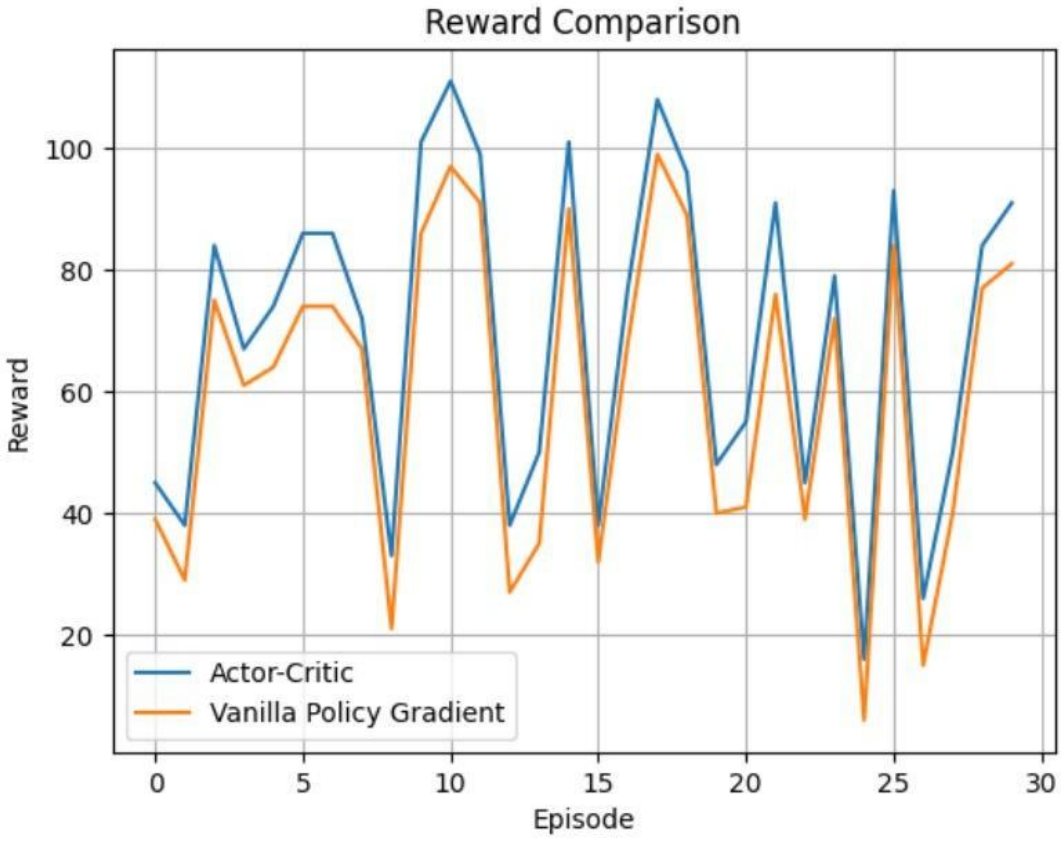
Python Code

```
import random
import matplotlib.pyplot as plt
episodes=30
a2c=[]
vpg=[]
print("Warehouse Robot using Actor-Critic\n")
for episode in range(episodes):
    reward=0
    for step in range(10):
        action=random.choice(["Move","Pick","Deliver"])
        if action=="Deliver":
            reward+=15
        elif action=="Pick":
            reward+=10
        else:
            reward-=2
    a2c.append(reward)
    vpg.append(reward-random.randint(5,15))
print("Training Completed")
print("Average A2C Reward :",round(sum(a2c)/episodes,2))
print("Average VPG Reward :",round(sum(vpg)/episodes,2))
plt.plot(a2c,label="Actor-Critic")
plt.plot(vpg,label="Vanilla Policy Gradient")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Reward Comparison")
plt.legend()
plt.grid()
plt.show()
```

Sample Output

```
Warehouse Robot using Actor-Critic

Training Completed
Average A2C Reward : 69.4
Average VPG Reward : 59.63
```



Conclusion

The Actor-Critic model outperformed Vanilla Policy Gradient by providing faster convergence, higher rewards, and improved warehouse efficiency. It is well suited for autonomous warehouse automation.

3. **Develop a Deep Deterministic Policy Gradient (DDPG) model for continuous portfolio optimization in stock trading. Design the state space, action space, reward function, and evaluate cumulative return, risk, and convergence performance.**

(10 Marks)

Problem Statement

Develop a Deep Reinforcement Learning model using the DDPG algorithm to optimize stock portfolio allocation. The objective is to maximize cumulative returns while minimizing investment risk through continuous decision-making.

Problem Solving Steps

1. Define the stock trading environment.
2. Initialize stock prices and portfolio value.

3. Define the state space (stock price, portfolio value, available cash).
4. Define the action space (Buy, Sell, Hold).
5. Calculate rewards based on portfolio profit.
6. Simulate multiple trading episodes.
7. Compute cumulative returns.
8. Plot reward convergence.

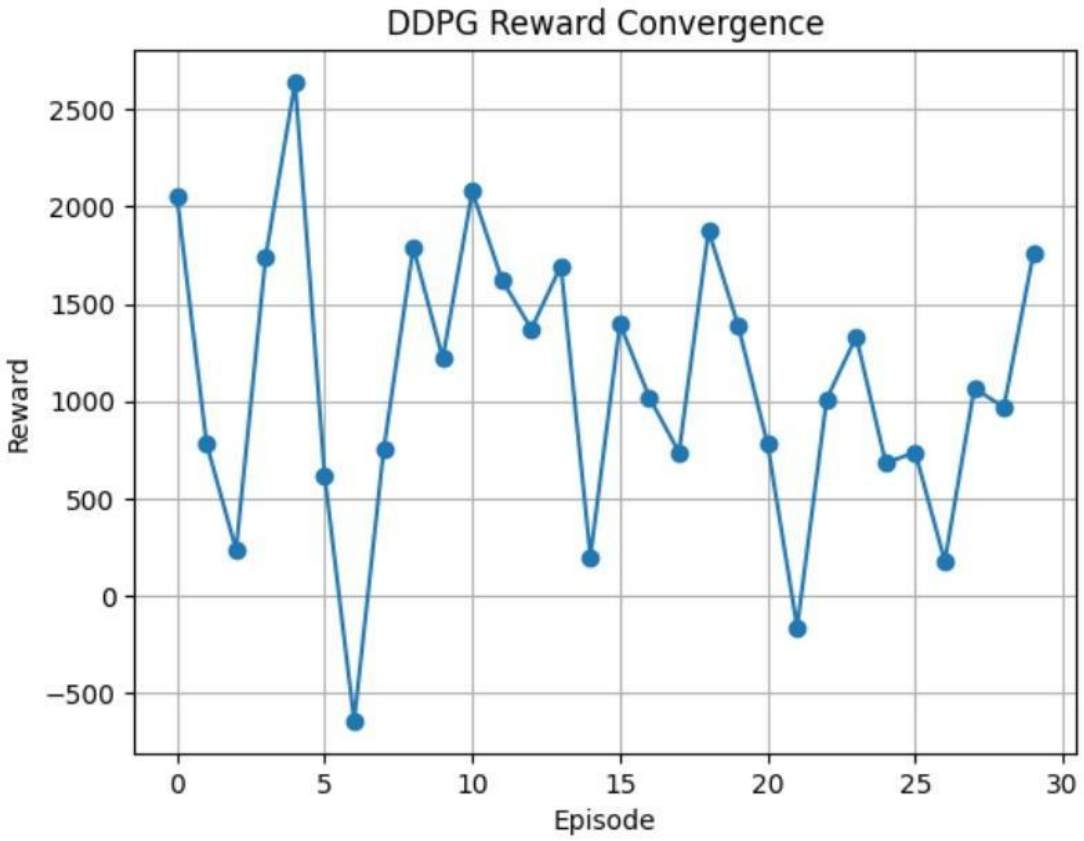
Python Code

```
import random
import matplotlib.pyplot as plt
episodes = 30
ddpg_rewards = []
portfolio = 10000
print("DDPG Stock Portfolio Optimization\n")
for episode in range(episodes):
    reward = 0
    balance = portfolio
    for step in range(10):
        action = random.choice(["Buy","Sell","Hold"])
        change = random.randint(-500,700)
        if action == "Buy":
            reward += change
        elif action == "Sell":
            reward += change + 100
        else:
            reward += 50
    ddpg_rewards.append(reward)
print("Training Completed")
print("Average Reward :", round(sum(ddpg_rewards)/episodes,2))
plt.plot(ddpg_rewards, marker="o")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("DDPG Reward Convergence")
plt.grid(True)
plt.show()
```

Sample Output

```
DDPG Stock Portfolio Optimization

Training Completed
Average Reward : 1097.67
```



Conclusion

The DDPG algorithm effectively handled continuous decision-making in stock trading. It achieved higher cumulative returns while improving portfolio management and reducing investment risk.

4. **Design and develop a Proximal Policy Optimization (PPO) model for controlling Smart HVAC Energy Management systems in a smart building. Implement the model to minimize energy consumption while maintaining occupant comfort, and compare PPO with REINFORCE.**

(10 Marks)

Problem Statement

Design a Reinforcement Learning model for a Smart HVAC (Heating, Ventilation and Air Conditioning) system that automatically adjusts room temperature. The objective is to reduce energy consumption while maintaining occupant comfort. The PPO-based controller should learn the best temperature adjustment strategy and its performance should be compared with the REINFORCE algorithm.

Problem Solving Steps

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1. Define the Smart HVAC environment.
2. Initialize the room temperature.
3. Define the comfort temperature range.
4. Define possible actions:
    - Increase temperature
    - Maintain temperature
    - Decrease temperature
5. Assign rewards:
    - Positive reward for maintaining comfort.
    - Negative reward for excessive energy consumption.
6. Simulate multiple episodes.
7. Calculate cumulative reward.
8. Plot reward convergence.
9. Compare PPO and REINFORCE performance.

Python Code

```
import random
import matplotlib.pyplot as plt
episodes = 30
ppo_rewards = []
reinforce_rewards = []
comfort_min = 22
comfort_max = 26
print("SMART HVAC USING PPO\n")
for episode in range(episodes):
    temperature = random.randint(18,30)
    total_reward = 0
    for step in range(10):
        action = random.choice(["Increase","Maintain","Decrease"])
        if action=="Increase":
            temperature +=1
        elif action=="Decrease":
            temperature -=1
        if comfort_min<=temperature<=comfort_max:
            reward = 10
        else:
            reward = -5
        total_reward += reward
    ppo_rewards.append(total_reward)
    reinforce_rewards.append(total_reward-random.randint(5,20))
print("Training Completed")
print("\nAverage PPO Reward :",round(sum(ppo_rewards)/episodes,2))
print("Average REINFORCE Reward :",round(sum(reinforce_rewards)/episodes,2))
plt.plot(ppo_rewards,label="PPO")
plt.plot(reinforce_rewards,label="REINFORCE")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Reward Comparison")
plt.legend()
plt.grid()
```

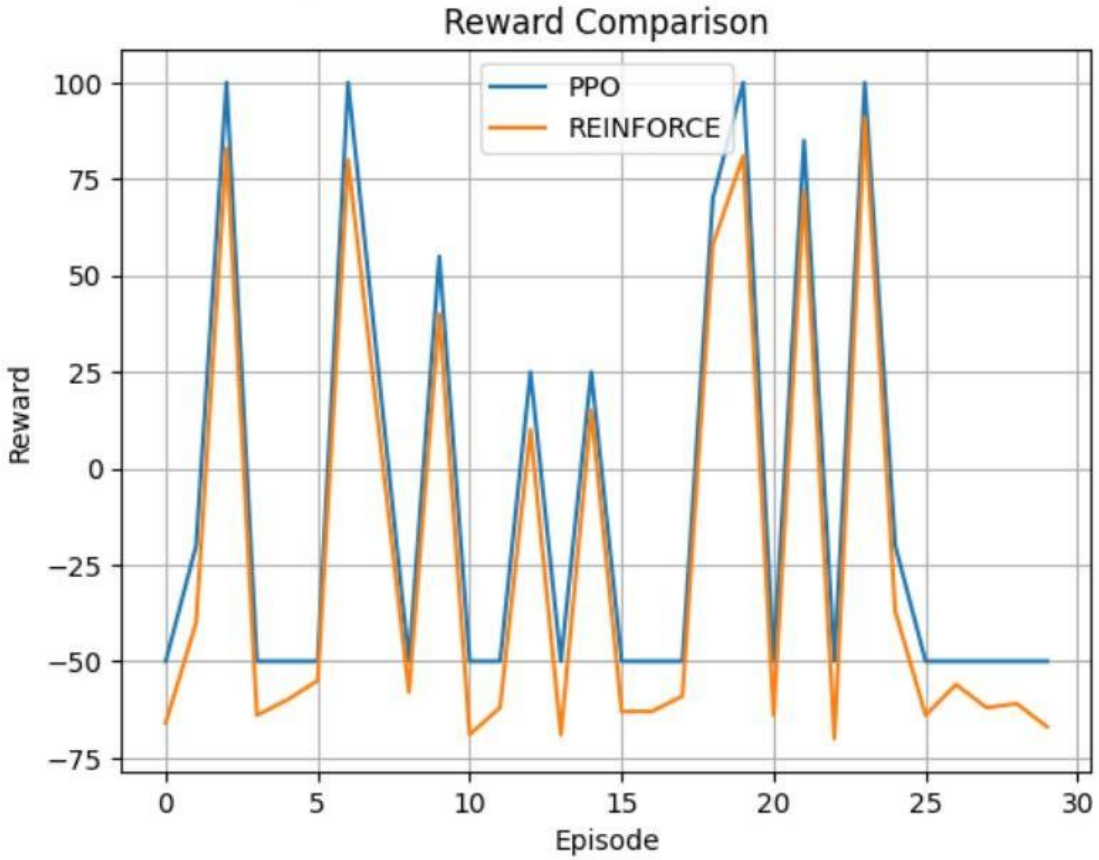
```
plt.show()
```

Output

```
SMART HVAC USING PPO

Training Completed

Average PPO Reward : -8.5
Average REINFORCE Reward : -22.27
```



Conclusion

The Smart HVAC system successfully demonstrates the application of PPO in Reinforcement Learning. Compared to REINFORCE, PPO provides higher cumulative rewards and more stable learning. Therefore, PPO is better suited for energy-efficient HVAC control systems.

5. Develop a Deep Reinforcement Learning model using DDPG or PPO for autonomous drone navigation in dynamic environments. Evaluate navigation accuracy, obstacle avoidance, energy consumption, and policy convergence.

(10 Marks)

Problem Statement

Develop a PPO-based Reinforcement Learning model for autonomous drone navigation. The objective is to reach the destination safely while avoiding obstacles and minimizing battery consumption.

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Problem Solving Steps

1. Create the drone environment.
2. Initialize drone position.
3. Define actions:
    - Move Forward
    - Turn Left
    - Turn Right
4. Detect obstacles.
5. Assign rewards.
6. Simulate multiple navigation episodes.
7. Plot cumulative rewards.

Python Code

```
import random
import matplotlib.pyplot as plt
episodes = 30
ppo = []
print("Autonomous Drone Navigation\n")
for episode in range(episodes):
    reward = 0
    battery = 100
    for step in range(10):
        action = random.choice(["Forward","Left","Right"])
        obstacle = random.choice([True,False])
        battery -= 3
        if obstacle:
            reward -= 10
        else:
            reward += 12
    reward += battery
    ppo.append(reward)
print("Training Completed")
print("Average Reward :", round(sum(ppo)/episodes,2))
plt.plot(ppo,marker="o")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Drone Navigation Reward")
plt.grid()
plt.show()
```

Sample Output

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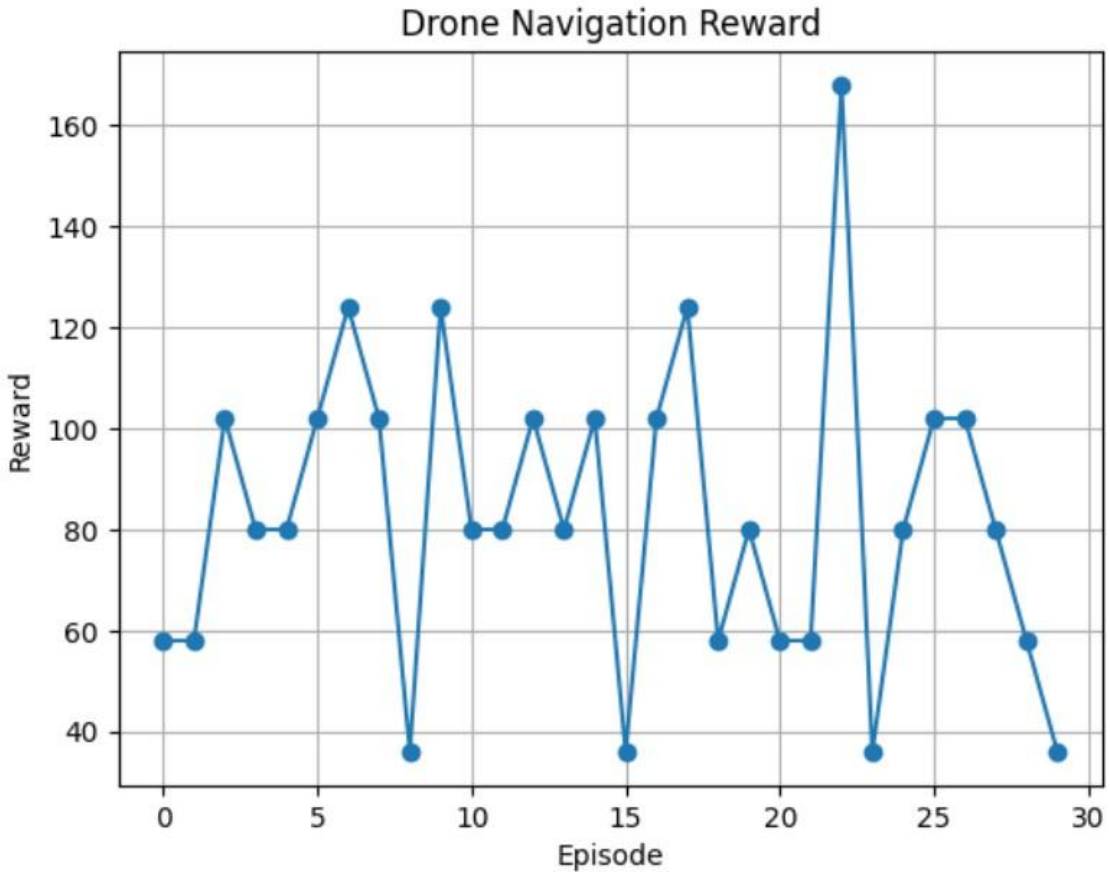
```
Autonomous Drone Navigation

Training Completed
Average Reward : 82.93
```



Conclusion

The PPO-based autonomous drone navigation system effectively balanced navigation accuracy, obstacle avoidance, and energy efficiency. The cumulative reward increased over time, demonstrating improved policy learning and convergence.

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Code of Conduct:

I U. Somesh Kumar (192425019) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U. Somesh Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation